Executive Project Preview Catalog

Phase 1 Architectural Baselines & Data Infrastructure Preview

ITAI 4160 / 5160: Databases and Cloud for Artificial Intelligence | **Date:** October 28, 2026

Course Instructor: Dr. Qiong Cheng | **Executive Guests:** Vanessa Vreeland & Ken Meyer

Presentation Schedule

Order	Team / Project	Project Title
1	FitWise AI	Personalized Gym Coaching & Algorithmic Workout Prescriptions
2	Financial Literacy Service	AI-Powered Market Context for Beginner Investors
3	ClassWizard	Intelligent Academic Scheduling & Course Planning Platform
4	S-Card	Personalized Spam Protection
5	CoursePilot	Autonomous Academic Degree Navigation & Advisory Assistant
6	RAEJ	LLM Translation for Languages Without Parallel Data
7	MCP Sentinel	From MCP Server Discovery to Explainable Security Risk Intelligence

Team Project Briefs (Early Baseline Work)

1. FitWise AI — Personalized Gym Coaching & Algorithmic Workout Prescriptions [Phase 1 Prototype]

Tech Stack: PostgreSQL + pgvector, PySpark (Bronze/Silver/Gold), FastAPI, React, Docker, AWS S3/EC2/RDS

Team Members: Josiah Duran (Team Leader), Raymundo Munoz Cuevas, Ali Chaaraoui (Team Leader), Essey Beraki

Project Brief: Learns from each user's training, sleep, and activity to prescribe workout splits/weights and forecast target goal dates using user-tuned XGBoost models and RAG-driven explanations.

Q&A: At what scale should we split workouts, wearable data, and vector embeddings into dedicated stores?

2. Financial Literacy Service: AI-Powered Market Context for Beginner Investors [Phase 1 Prototype]

Tech Stack: React, TypeScript, Python 3.12, FastAPI, PostgreSQL 16, pgvector, OpenAI Embeddings, Docker Compose

Team Members: Josh Marsico (Team Leader), Jayden Parsons, Himaja Goda, Michael Simms

Project Brief: Connects market data with financial news to explain market trends and financial terms to beginner investors using citation-backed AI summaries.

Q&A: What governance, audit logging, and human-review controls should be in place before an AI system presents financial-news explanations to customers?

3. ClassWizard — Intelligent Academic Scheduling & Course Planning Platform [Phase 1 Prototype]

Tech Stack: PostgreSQL, pgvector, Neo4j, Docker, AWS (EC2, S3, Lambda), FastAPI, Open-Source LLMs

Team Members: Akshat Kumar, Caleb Richardson (Team Leader), Jackson Redman, Scott Do

Project Brief: A career-aware, prerequisite-focused planning system that aligns university course offerings with student career goals and industry skill requirements.

Q&A: How can organizations protect sensitive data when integrating information from internal systems into data analytics and AI applications?

4. S-Card Personalized Spam Protection [Phase 1 Prototype]

Tech Stack: Python / scikit-learn, ONNX Runtime, Streamlit, PostgreSQL, Docker, AWS EC2, S3

Team Members: Smriti Bhemireddy, Connor Kelly, Andrew Gallegos Garcia (Team Leader), Rithvik Gottumukkala (Team Leader), Dash Doubleday

Project Brief: Filters messages using a portable personal profile and local AI without requiring a name or email, enabling on-device, privacy-preserving spam protection for SMS and email.

Q&A: 1) How should organizations balance response time, device battery life, data privacy, and cost when deciding whether an AI application should run locally or use cloud-based inference?
2) What level of false positives would a bank consider acceptable before piloting this system with customers, and what safeguards and remediation processes should be in place when errors occur?

5. CoursePilot — AI Academic Advising Assistant [Phase 1 Prototype]

Tech Stack: React / Next.js, Python / FastAPI, Amazon RDS (PostgreSQL + pgvector), Amazon S3, Docker, LLMs

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Team Members: Mujtaba Ali Khan, Cayman Powell (Team Leader), Alexis Arenas-Villalba

Project Brief: An autonomous academic advising assistant providing transparent, plain-language answers to degree requirement questions with traceable source citations.

Q&A: What's the best way to balance managed cloud services with a containerized deployment without having to sacrifice student uploads and with the ability to keep source traceability?

6. RAEJ: LLM Translation for Languages Without Parallel Data [Phase 1 Prototype]

Tech Stack: Python, LangChain, PostgreSQL + pgvector, LM Studio, Ollama, OpenAI API, AWS EC2, Docker Compose

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Team Members: Alec Pike, Ryan Brown, Eric Medlock (Team Leader), Justin MacCreery

Project Brief: Evaluates whether an LLM can translate reliably without parallel text using dictionary and grammar references alone via retrieval-augmented generation.

Q&A: As context windows grow from thousands to millions of tokens, is it still worth the complexity to use RAG? How do you evaluate and decide?

7. From MCP Server Discovery to Explainable Security Risk Intelligence [Phase 1 Prototype]

Tech Stack: PostgreSQL (JSONB), Python, Docker, AWS (Candidate), MCP Registries, OSV.dev / deps.dev APIs

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Team Members: Sana Shaik, Sahir Shaik, Shreem Dixit, Dias Utebekov (Team Leader)

Project Brief: Delivers security visibility and risk intelligence for Model Context Protocol (MCP) servers used by AI agents to connect databases, APIs, and business apps.

Q&A: When a third-party MCP server has no known vulnerabilities but hasn't been fully assessed, what evidence and safeguards should a bank require before approving it?